



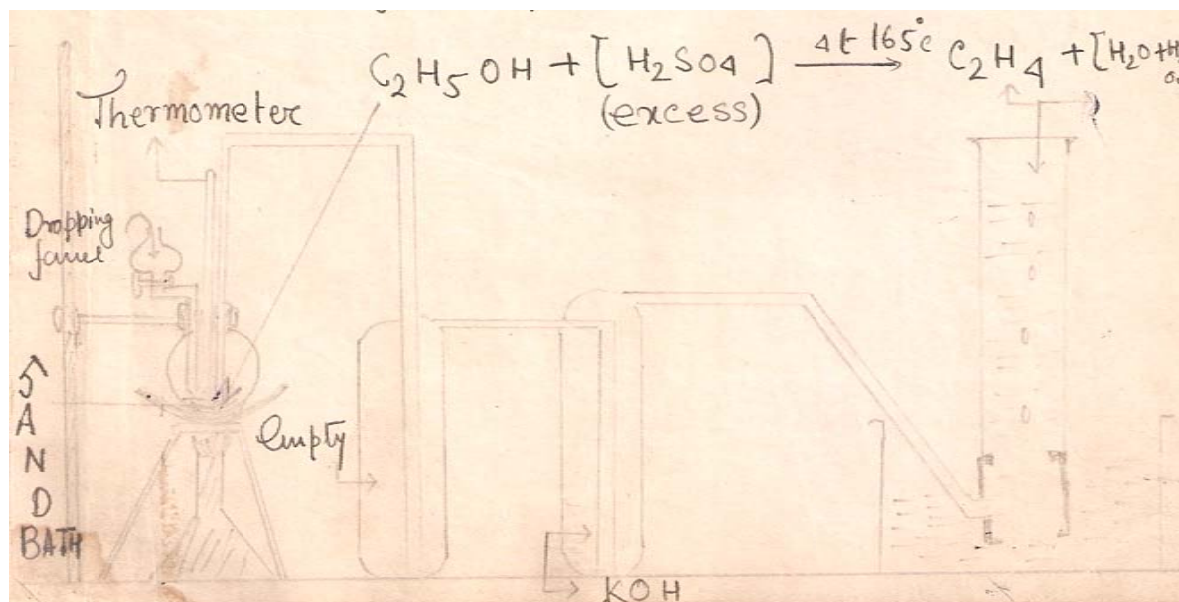
How is Ethylene gas prepared in the laboratory in fairly pure form. What are the other methods of its preparation? Why Ethylene is more reactive than Ethane? Discuss the properties of ethylene and mention the uses of it.

Preparation of Ethylene -

1. Theory → When a mixture of excess of con. H_2SO_4 and C_2H_5OH ^{is heated} at nearly $165^\circ C$ we get ethylene.

$$C_2H_5OH + [H_2SO_4] \xrightarrow{at\ 165^\circ C} C_2H_4 + [H_2O + H_2SO_4]$$

excess





Procedure — In a round bottle flask about 100°C of con. H_2SO_4 is mixed with about 50 c.c of $\text{C}_2\text{H}_5\text{OH}$ with the help of dropping funnel. The flask is fitted with a thermometer to record the temperature $165^{\circ} - 170^{\circ}\text{C}$. It is also fitted with a delivery tube which is connected with an empty washing bottle and then connects with another washing bottle containing KOH solution. It is then connected to a gas jar as shown in the figure. The flask is clamped and heated over sand bath.

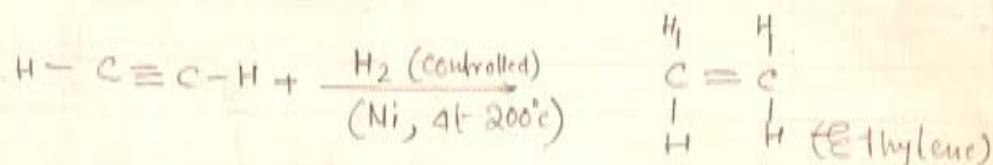
On heating the mixture between $165^{\circ} - 170^{\circ}\text{C}$ we get ethylene gas. This gas is collected by the downward displacement of water.

Precaution — The heating should be done over sand bath in between $165^{\circ}\text{C} - 170^{\circ}\text{C}$ otherwise different product may be obtained. An empty washing bottle is used as anti suction device.

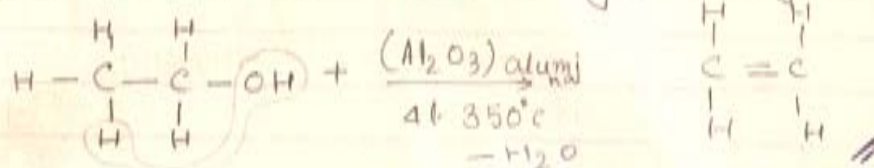
Purification — This gas is purified by passing through KOH solution taken in washing bottle which absorb CO_2 and SO_2 present as impurities.

Other method of preparation

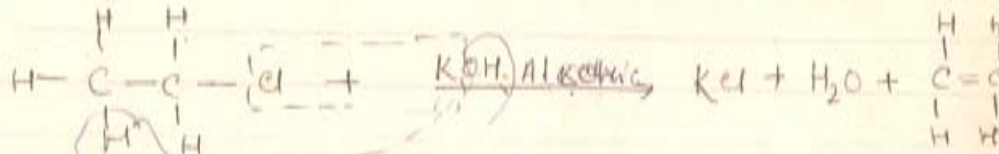
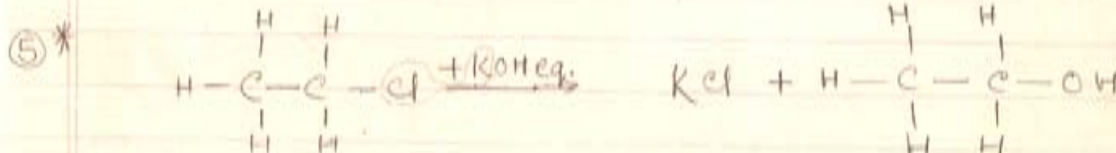
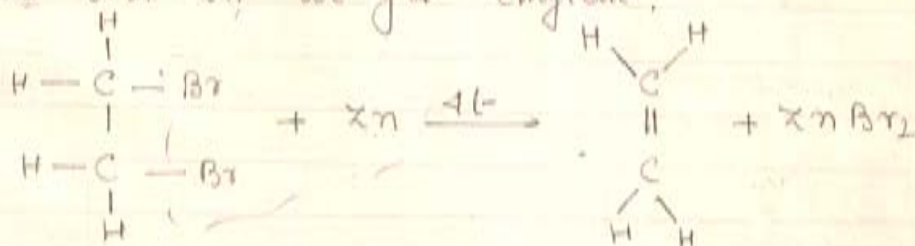
When Acetylene gas is heated with hydrogen in the presence of nickel at about 200°C , in a controlled way we get ethylene.



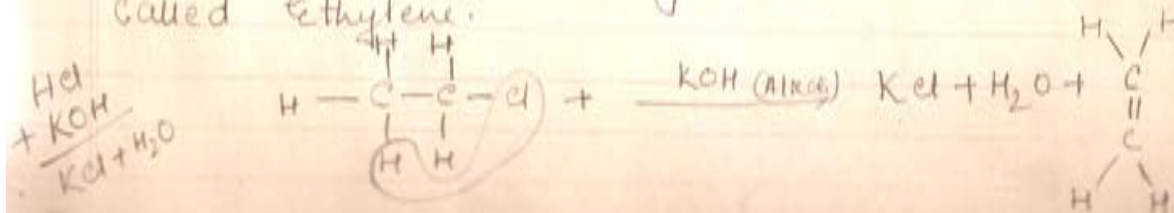
3. When the vapour of $\text{C}_2\text{H}_5\text{OH}$ is passed through Al_2O_3 heated at 350°C we get Ethylene.



④ ^{W.H} When Ethylene Bromide / Ethylene di Bromide is heated with Zn , we get Ethylene.

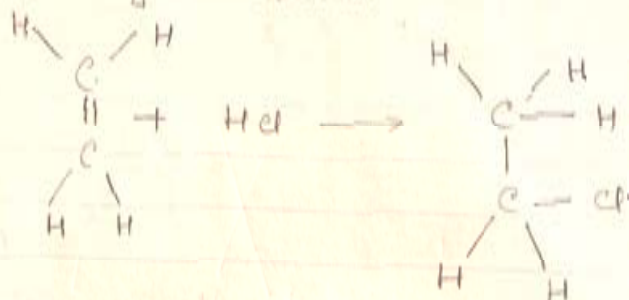


When Ethyl Chloride / Ethyl halide is heated with Alcoholic KOH , we get unsaturated compound called Ethylene.

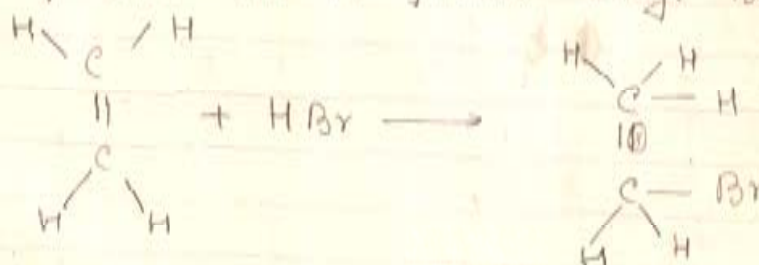




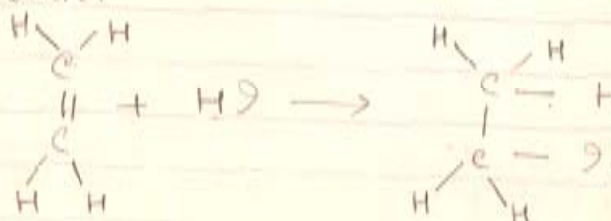
4. With $\text{HCl} \rightarrow$ When Ethylene is treated with HCl , we get Ethyl chloride



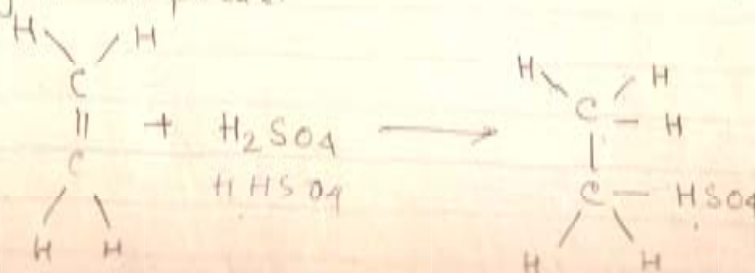
5. Similarly with HBr It forms Ethyl Bromide



6. With HI — It reacts with HI and made Ethyl Iodide.

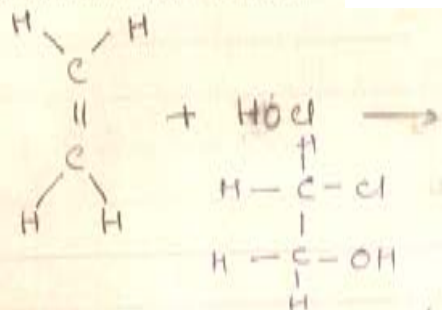


7. With $\text{H}_2\text{SO}_4 \rightarrow$ When Ethylene is passed through con. H_2SO_4 due to addition reaction we get Ethyl hydrogen Sulphate.

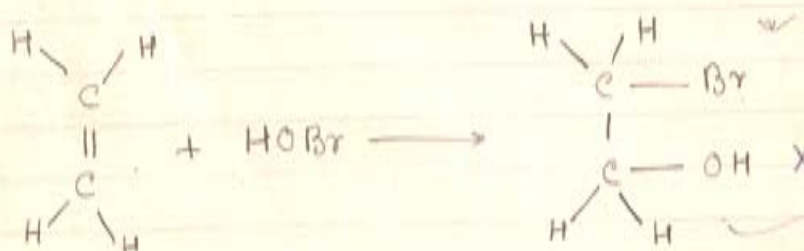
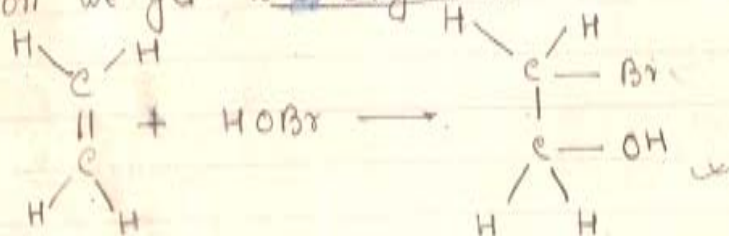




8. With hypochlorous acid (HOCl) — When Ethylene gas is passed through hypochlorous acid, due to addition reaction **Chloro hydrine** is formed

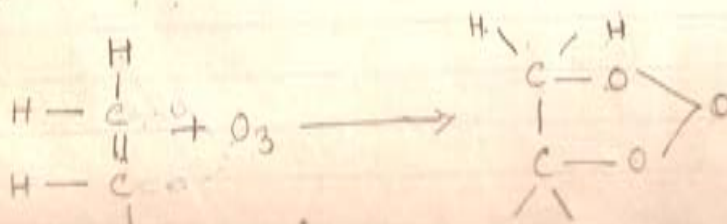


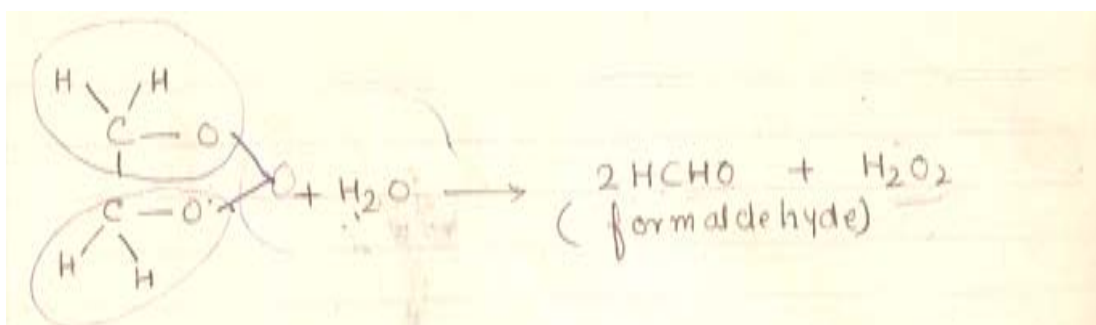
9. With hypobromous acid (HOBr) — When Ethylene gas is passed through HOBr due to addition reaction we get Bromohydrine



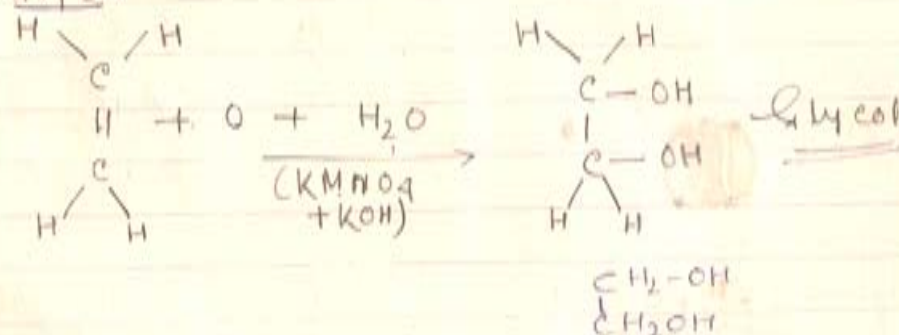
⑩ W.H

- With Ozone (O₃) — When Ethylene reacts with Ozone giving Ethylene ozonide, a ring compound which with a little water breaks into formaldehyde and H₂O₂





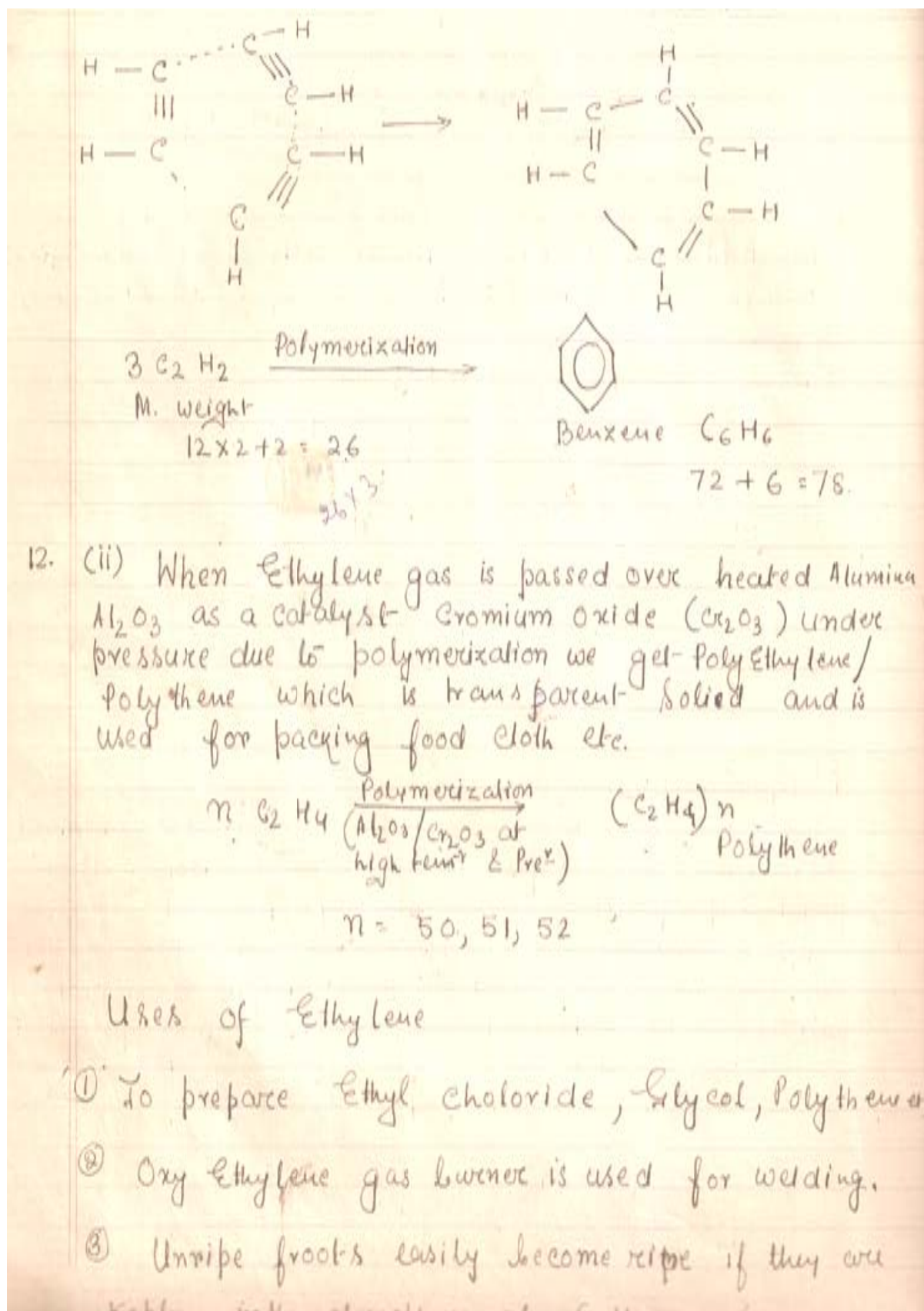
With KMnO_4 solution - with a little KOH (alkalified KMnO_4) It reacts to form a sweet liquid called Glycol



Polymerization

When two or more organic compound combine to form a new compound whose chemical and physical properties are completely different than original organic compounds. The molecular weight of the new compound is the multiple of the molecular weight of the original compound. Due to polymerization no water molecule and any other substance is removed. i.e

(i) When Acetylene gas is passed through a red hot copper tube, due to polymerization Benzene is formed.





Explain why Ethylene is more reactive than Methane

In Ethylene there is a double bond between Carbon atoms due to this double bond much strain is expected according to Baeyer strain theory. So it is highly reactive and addition reactions are very common.

On other hand Ethane has a single bond between carbon atoms so there is no strain. So it does not give addition reaction. However it gives only substitution reaction.

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